To solve this exercise, we remember that if a variable x can take any real value, it can be replaced by two non negative artificial variables denoted by x^+ and x^-, such that $x = x^+ - x^-$. We also recall that the absolute value of x is defined as

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

In our case, we have that

$$|x_1 - x_2| = \begin{cases} x_1 - x_2 & \text{if } x_1 \geq x_2, \\ x_2 - x_1 & \text{if } x_1 < x_2. \end{cases}$$

Since x_1 and x_2 are non negative real numbers, the difference $x_1 - x_2$ can take any real value.

Let us study the absolute value by examining the two cases:

- If $x_1 - x_2 \geq 0$, we can define the non negative quantity

$$y^+ = \begin{cases} x_1 - x_2, \text{ if } x_1 - x_2 > 0, \\ 0, \text{ otherwise.} \end{cases}$$

- If $x_1 - x_2 < 0$, we can define the positive quantity

$$y^- = \begin{cases} x_2 - x_1, \text{ if } x_1 - x_2 < 0, \\ 0, \text{ otherwise.} \end{cases}$$

Consequently, the absolute value of the difference can then be written as

$$|x_1 - x_2| = y^+ + y^-.$$

Furthermore, if we impose that $y^+ \geq 0$ and $y^- \geq 0$, we have that our minimization problem can be written as

$$\min_{x_1, x_2, y^+, y^-} (y^+ + y^-)$$

subject to

$$\begin{aligned} y^+ &\geq x_1 - x_2, \\ y^- &\geq x_2 - x_1, \\ x_1 &\geq 0, \\ x_2 &\geq 0, \\ y^+ &\geq 0, \\ y^- &\geq 0. \end{aligned}$$

Note that this formulation is not strictly equivalent to the original one for all feasible solutions. For instance, the feasible solution $x_1 = 0$, $x_2 = 0$, $y^+ = 1$, $y^- = 1$ has objective value 2 in the transformed problem, and 0 in the original one.

But it is equivalent at the optimal solution. Denote the optimal solution of the original problem (x_1^*, x_2^*).

- If $x_1^* - x_2^* > 0$, the lowest possible value for y^+, denoted by $(y^+)^*$, is $x_1^* - x_2^*$ (because of the constraint $y^+ \geq x_1 - x_2$), and the lowest possible value for y^-, denoted by $(y^-)^*$, is 0 (because of the constraint $y^- \geq 0$). Therefore, the objective function of the transformed problem at the optimal solution is

$$(y^+)^* + (y^-)^* = x_1^* - x_2^* + 0 = |x_1^* - x_2^*|.$$

- If $x_1^* - x_2^* < 0$, for similar reasons, we have $(y^+)^* = 0$ and $(y^-)^* = x_2^* - x_1^*$. Therefore, the objective function of the transformed problem at the optimal solution is

$$(y^+)^* + (y^-)^* = 0 + x_2^* - x_1^* = |x_1^* - x_2^*|.$$

- If $x_1^* - x_2^* = 0$, we have $(y^+)^* = 0$ and $(y^-)^* = 0$. Therefore, the objective function of the transformed problem at the optimal solution is

$$(y^+)^* + (y^-)^* = 0 + 0 = |x_1^* - x_2^*|.$$